A Case Study of Predicting the Class of the Unknown Viruses of ssRNA positive-strand in NCBI

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Abstract

The purpose of this paper is to predict the class label of unknown virus in the class of ssRNA positive-strand in NCBI. Each virus (sequence) was presented as one vector via representative patterns. The representative patterns are distinct patterns with biased distribution of the frequency among classes. We use WithinCoverage *and* PatternEntropy *as thresholds to select representative patterns. The WithinCoverage of one pattern is the percentage of the sequences within one class in which that pattern appears. The PatternEntropy is the entropy of term frequency of one pattern across different classes. Note that the length of these patterns are variable. We experimented with the ssRNA positive-strand viruses in NCBI in which consisted of* 33 *classes with* 396 *viruses as training data and the unknown class with* 8 *viruses as testing data. We adapted the linear classifier with Rocchio algorithm to measure the relationship between the existed class and the unknown viruses. Experimental results showed the class achieved the highest value of cosine similarity for each unknown virus.*
Keywords: *representative patterns, classification.*

1 Introduction

The DNA sequences of entire genomes are being determined at a rapid rate. As more and more whole genomes are available, there is a need for new methods to compare large sequences and transfer biological knowledge from annotated genomes to related new ones. These changes in genome sequencing strategies have created a strong need for new methods to compare genomic sequences. There are sequences alignment methods for sequence comparison, like Smith-Waterman algorithm [17], FASTA algorithm [13] and BLAST algorithm [4], to measure the relationship between different sequences. However, these methods usually suffer from the limitations of the computational time and memory available to handle large amount of DNA sequences data.

The purpose of this paper is to predict the class label of unknown virus in the class of ssRNA positive-strand in NCBI [3]. We present each virus as one vector using representative patterns such that many machine learning algorithms in vector space model [14] could be applied to those bioinformatics problems [9], such as inter or intra-chromosomal comparison. The representative patterns according to the classification, determined by domain specific experts, are distinct patterns with biased distribution of the frequency among classes.

We select representative patterns from the significant patterns. First of all, we adapted our Chinese term extraction approach [18, 22] for extracting the significant patterns from DNA sequences. Note that the length of these patterns are variable. To obtain the distribution of the frequency of each pattern among classes, first we sort all of the suffix strings [7] from the sequences. Then, we scan the sorted suffix strings in lexicographic order and extract the longest common prefix and, with our significant term extraction approach, calculate the statistics of the frequency distribution among classes. The patterns extracted from above processes are candidate significant patterns with right boundary verification. We do the same processes with the reversed strings of the original sequences to have patterns with the left boundary verification. The significant patterns are candidate significant patterns that passed both right and left boundary verification. Finally, the representative patterns are selected from those significant patterns with biased frequency distributions among classes.

After extracting the significant patterns, we use *WithinCoverage* and *PatternEntropy* as thresholds to select representative patterns from the significant patterns. The WithinCoverage of one pattern is the percentage of the sequences within one class in which that pattern appears. The PatternEntropy is the entropy of term frequency of one pattern across different classes. Intuitively, we select the patterns with the value of WithinCoverage as high as possible such that these patterns appear in most of the sequences within the same class. Therefore, these patterns selected are representative of that class. On the other hand, we select the patterns with the value of PatternEntropy as low as possible such that the distributions of term frequency of these patterns among classes were biased. That is, these patterns could be applied to distinguish one class from the others. After selecting the representative patterns, we can present a sequence (a virus) as one vector. To determine the weighting for each dimension (representative pattern) in one vector, we use the combination of the length of representative pattern and the entropy of term frequency distribution of that pattern among the classes existed.

In this paper, for simplicity, we have linear classifier using Rocchio algorithm [8] to predict the class label of unknown virus, although there are many well-known classifers [10]. We experimented with the ssRNA positive-strand viruses [3] (Revised: March 5, 2004.) from NCBI [1]. There were 33 classes which consisted of 396 viruses as training data and the unknown ssRNA positive-strand viruses which consisted 8 viruses as testing data. Experimental results showed the class achieved the highest value of cosine similarity for each unknown virus.

Because there were some classes containing few virus in the training set, we didn't have the measure of precision and recall [14] in classification to evaluate the performance of virus classification, but only predicted the unknown virus. We are looking for the classes containing more sequences to further investigate our approach for sequence classification [11, 2] such that make our experiments more robust in the future.

The remainder of this paper is organized as follows. Section 2 describes the processes of extracting representative patterns. Section 3 gives the weighting method of representative patterns for each sequence. Section 4 reviews the linear classifier with Rocchio algorithm. Section 5 gives experimental results. Section 6 gives conclusions and discussions.

2 Extracting Representative Patterns

We adapted our Chinese term extraction approach [18, 22] for extracting the representative patterns to distinguish one class from the others. Note that the length of these patterns are variable. The process of representative patterns extraction consists of three steps: (i) sorting the suffix strings of genomic sequences, (ii) determining the boundaries of significant patterns and (iii) selecting representative patterns from significant patterns. First, we sorted the suffix strings of genomic sequences. Then, we have significant patterns which pass both right boundary and left boundary verification by scanning the suffix strings lexicographically and extracting the longest common prefix from adjacent strings. Finally we select representative patterns from significant patterns according to two parameters. One is the percentage of species within one class that pattern appeared and another is the entropy of the pattern frequency distribution among classes. Intuitively, we aim to extract those representative patterns which appear in the species within the same class as many as possible, and whose frequency distributions among classes were biased such that they could be applied to distinguish different classes with proper weighting method. We described these three steps in section 2.1, section 2.2 and section 2.3, respectively.

2.1 Sorting Suffix Strings

The suffix strings of the sequences from the genome of each species are sorted such that we can scan these strings in lexicographical order to verify the boundaries of significant patterns and, at the same time, to have the statistics of attributes related to those sequences. Note that we assume the maximum length of patterns is 50 in this paper. As shown in Fig. 1, for example, there are suffix strings of one sequence "TATAAAAGGAT". Note that the reversed sequence of "TATAAAAGGAT" is "TAGGAAAATAT". The suffix strings of the sequences and those of the reversed sequences are individually sorted such that all the suffix strings are in lexicographic order and are used to determining the right or left boundaries of significant patterns which was de-

(1) TATAAAAAGGAT
(2) ATAAAAAGGAT
(3) TAAAAAGGAT
(4) AAAAAGGAT
(5) AAAAGGAT
(6) AAAGGAT
(7) AAGGAT
(8) AGGAT
(9) GGAT
(10) GAT
(11) AT
(12) T

Figure 1: Suffix strings of sequence "TATAAAAAGGAT"

fined in section 2.2. A partial sorted suffix strings, for example, are shown in Fig. 2, where the value of "SID", for simplicity, are 1 for the identifier of one sequence, and the value of "Position" represents the starting position of suffix string in that sequence.

2.2 Determining the Boundaries of Significant Patterns

First of all, we give the definition of significant pattern [21] as following.

Definition 1 Significant Pattern
Let w be a string. For any character x, let $Porb(wx|w)$ be the probability that w is followed by x, and $Porb(xw|w)$ be the probability that w is preceded by x. The string w passes the right boundary verification if $Porb(wx|w) < \theta_1$ for all x, and passes the left boundary verification if $Porb(xw|w) < \theta_2$ for all x, where θ_1 and θ_2 are threshold determined manually. The probability $Porb(wx|w)$, resp. $Porb(xw|w)$, is estimated by $\frac{TF(wx)}{TF(w)}$, resp. $\frac{TF(xw)}{TF(w)}$, where $TF(y)$ is the term frequency of string y. The string w is identified as a *significant term* if it passes both right and left boundary verifications and its term frequency $TF(w)$ exceeds a threshold, δ. This study simply sets $\theta_1 = \theta_2 = 1$ and $\delta = 2$. That is, w is identified as a significant term if it has at least two distinct succeeding and preceding characters, and if $TF(w)$ exceeds 2.

The sorted suffix strings are scanned lexicographically to extract the longest common prefix of adjacent strings yielded candidate significant patterns with right and left boundary verification via our term extraction approach in [18, 22]. As shown in Fig. 1, the pattern "TATAAA", for example, is a significant pattern which pass both right and left boundary verification. However, As shown in Fig. 1 (a), the pattern "TATAAAGT" with dotted rectangle, for example, didn't pass right boundary verification because it was always covered by the pattern "TATAAAGTT". Similarly, as shown in Fig. 1 (b), the pattern "AAATATTAATATT" with dotted rectangle, for example, didn't pass left boundary verification because it was always covered by the pattern "AAATATTAATATTA".

2.3 Selecting Representative Patterns from Significant Patterns

The representative patterns are used to distinguish one class from the others. We use *WithinCoverage* and *PatternEntropy* as thresholds to select representative patterns from the significant patterns. We define WithinCoverage and PatternEntropy in the following.

Definition 2 WithinCoverage
Let the WithinCoverage WC_{P_i,C_j} of pattern P_i respect to class C_j be $\frac{S_{P_i,C_j}}{|C_j|}$, where S_{P_i,C_j} is the number of sequences within class C_j that contain pattern P_i, and $|C_j|$ is the number of sequences within class C_j.

Definition 3 PatternEntropy
Let the PatternEntropy $PE(P_i)$ of pattern P_i be $\frac{\sum_{k=1}^{k=|C|} Prob(P_{i,k}) * log_{|C|} Prob(P_{i,k})}{log|C|}$, where the $|C|$ is the number of classes and the probability $Prob(P_{i,j})$ is $\frac{TF_{i,j}}{\sum_{k=1}^{k=|C|} TF_{i,k}}$ and $TF_{i,j}$ is the term frequency of pattern P_i in class C_j. Note that $0 \leq PE(P_i) \leq 1$

The WithinCoverage of one pattern is the percentage of the sequences within one class in which that pattern appears. The PatternEntropy is the entropy of term frequency of one pattern across different classes. Intuitively, we select the patterns with the value of WithinCoverage as high as possible such that these patterns appear in most of the sequences within the same class. Therefore, these patterns selected are representative of that class. On the other hand, we select the patterns with the value of PatternEntropy as low as possible such that the distributions of term frequency of these patterns among classes were biased. That is, these patterns could be applied to distinguish one class from the others.

(SID :Position) Suffix String of Sequences

⋮

1	7801	TAGTTTTTGATGGCAAGTCCAAATGCGACGAGTCTGCTTCTAAGTCTGCT
1	5877	TATAAAAAGGATAATGCTTACTATACAGAGCAGCCTATAGACCTTGTACC
1	4880	TATAAAAGTGTTCACAACTGTGGACAACACTAATCTCCACACACAGCTTG
1	109	TATAAACAATAATAAATTTTACTGTCGTTGACAAGAAACGAGTAACTCGT
1	5823	TATAAACTCGATGGAGTTACTTACACAGAGATTGAACCAAAATTGGATGG
1	1718	TATAAAGAGTCTTGATTACAAGTCTTTCAAAACCATTGTTGAGTCCTGCG
1	21040	TATAAAGCAAAAACTAGCCCTGGGTGGTTCTATAGCTGTAAAGATAACAG
1	4392	TATAAAGGAATTAAAATTCAAGAGGGCATCGTTGACTATGGTGTCCGATT
1	18413	TATAAAGGCTTGCCCTGGAATGTAGTGCGTATTAAGATAGTACAAATGCT
1	1773	TATAAAGTTACCAAGGGAAAGCCCGTAAAAGGTGCTTGGAACATTGGACA
1	18043	TATAAAGTTCAAGACTGAAGGATTATGTGTTGACATACCAGGCATACCAA
1	22787	TATAAATATAGGTATCTTAGACATGGCAAGCTTAGGCCCTTTGAGAGAGA
1	26988	TATAAATTAAATACAGACCACGCCGGTAGCAACGACAATATTGCTTTGCT
1	5619	TATAAATTACAGCAAGGTACATTCTTATGTGCGAATGAGTACACTGGTAA
1	22703	TATAAATTGCCAGATGATTTCATGGGTTGTGTCCTTGCTTGGAATACTAG
1	10269	TATAAATTTGTCCGTATCCAACCTGGTCAAACATTTTCAGTTCTAGCATG
1	9810	TATAACAAGTACAAGTATTTCAGTGGAGCCTTAGATACTACCAGCTATCG

⋮

(a) sorted suffix strings

(SID :Position) Suffix String of Sequences

⋮

1	12550	AAATAGTTTTGTGAGATGCGCGCGTGCGCGTCCATACTAAGATGATGTAA
1	11687	AAATATAGTTGCGACTCCACACATCCACGGACACATCCTACTTCTGGTCA
1	2742	AAATATCAAAGGTTATGCCATCGCCAACATACGTCGTTTTGGACTTAGTC
1	27957	AAATATCAATGGCGTCCTGAGTTGTTACCAAAACTTTCTGAACATTAGTT
1	28012	AAATATCACAGTTATTTCCGTGAACATCTTCGTCTCTTTCTACGGTTTTA
1	24850	AAATATCAGAATTGGAGGGCGTCCCTATTCTCTGAGAAATCAAACAGTTC
1	19461	AAATATGAACCCACAGAATCCCAATCTTCATAGTTGAAATTCGGATTCGT
1	25338	AAATATGAATGCAACCTACCAACGGTAATACCGAGATTGTAGGTACGTAT
1	29621	AAATATGACGCATCCACGTGATCCGTACGTCGGCTCGCTGTCGATGTGTC
1	24111	AAATATGAGTCGTCCACCACGTCTGTAGTATTGTTTTCTTCTGAGAACAA
1	23907	AAATATGCTGTGTCCGAACTACCAACATCACATTCTACAAAGGAACATCT
1	11317	AAATATGTACTCACCATATTCTACAAATTTGACCAGTGGACCACCTCCAA
1	6943	AAATATTAATATTAATGGTCAACTTCATCGTAGTTACAAGGATCATAAGG
1	7027	AAATATTAATATTAGTCGTTATTGTGGTCAAACAGGACCGCGATAAACAG
1	23853	AAATATTATGGGTAGGTTAAAACCAAGTTAGAGACACATTCATTGAGGTA
1	8690	AAATATTTAGGTGTGTCTATTCACTTTTTGGGAAGAAATCTCAGTAAGAG
1	24886	AAATCAAACAGTTCACTTTCTTGGAGTGGCAGTTCTACTTTGAGCTGCCC

⋮

(b) sorted suffix strings(reversed)

Figure 2: Sorted Suffix Strings

3 The Weighting of Representative Patterns for Each Sequence

In this study, we have the weighting w_j of representative pattern RP_j for the sequence W as $w_j = wtf(RP_j) \cdot length(RP_j) \cdot \frac{1}{PE(RP_j)+0.001}$, where $wtf(RP_j)$ is the term frequency of the pattern RP_j in the sequence W, and the PatternEntropy $PE(RP_j)$ be the entropy [10] of the pattern RP_j among classes, and the value 0.001 is set to avoid the value of PatternEntropy being zero. Therefore, we have each sequence $W = (w_1, w_2, \cdots, w_n)$ be represented as a vector, where n is the number of representative patterns selected.

4 Linear classifer using Rocchio Algorithm

Linear classifier is a simple approach for classification [8]. The main idea of linear classifier is to construct a feature vector as one representative for each class (category). For each class C_i, linear classifier computes prototype vector $G_i = (g_{i,1}, \ldots, g_{i,n})$, where n is the dimension of the pattern space and each element $g_{i,j}$ corresponds to the weight of the pattern P_i of G_i. The elements in the vector G_i are learned from positive examples tuned by negative examples. Positive examples are those documents belonging to that class while negative examples are those documents not belonging to that class. To classify a request sequence X, we compute the cosine similarity between X and each prototype vector G_i, and assign to X the class whose prototype vector has the highest degree of cosine similarity with X. Cosine similarity is defined as follows:

$$CosSim(X, G_i) = \frac{\sum_{j=1}^{n} x_j \cdot g_{i,j}}{\sqrt{\sum_{j=1}^{n} x_j^2}\sqrt{\sum_{j=1}^{n} g_{i,j}^2}}$$

In this study, we use the Rocchio algorithm [15] to construct the representative G_i for class C_i. Let P and N be the set of positive and negative examples with respect to class C_i in the training corpus. $|P|$ and $|N|$ are the number of examples in P and N, respectively. The prototype vector G_i is defined as follows:

$$G_i = \frac{\sum_{W \in P} W}{|P|} - \eta \frac{\sum_{W \in N} W}{|N|}$$

where η is the parameter that adjusts the relative impact of positive and negative examples. In this study, we choose $\eta = 0.25$ according to our experience in [19].

5 Experimental Results

We experimented with the ssRNA positive-strand viruses [3] in NCBI [1]. As shown in Table 1, we had 33 classes which consisted of 396 viruses as training data and, as shown in Table 2, the unknown class which consisted of 8 viruses as testing data. Note that we have one sequence represent for one virus.

First of all, we selected 19465 representative patterns from the significant patterns extracted from the training data. The representative patterns were the significant patterns which achieved the value of WithinCoverage over 95% for one class and the value of PatternEntropy under 0.2. The value of 95% and 0.2 were determined manually according to the average of non-zero items in vectors. We used these patterns to present the sequences of viruses as vectors. The average of non-zero items in one vector for training data and testing data are 105.5 and 35.5, respectively. The minimum and maximum length of representative patterns are 9 and 50, respectively. For simplicity, we only showed the distribution of the length of representative patterns whose lengths ranged from 9 to 30 in Figure 5. As shown in Table 3, we had the class achieved the highest value of cosine similarity for each unknown virus. For example, the value of cosine similarity of the "Botrytis virus F" and class "HepatitisE_likeviruses" was 0.316871.

6 Conclusions and Discussions

In this paper we predicted the unknown virus of the ssRNA positive-strand [3] in NCBI [1]. We extracted representative patterns from virus and presented the sequences as vectors such that makes the linear classifier for predicting unknown virus possible. Note that the length of representative patterns are variable.

We are looking for more sequences which are well classified and contain more sequences to further investigate our approach for DNA sequence classification such that make our experiments more robust in the future[2]. DNA sequence classification is an effective aid of the comprehension, summarization and compression of DNA sequences [11]. There are studies about the feature selection for sequence classification [6, 12, 20, 5].

Table 1: Training Data : ssRNA positive-strand viruses, no DNA stage(Revised: March 5, 2004.)

	Class	# of Virus	TotalLength(bp)
1	Astroviridae	6	41148
2	Barnaviridae	1	4009
3	Benyvirus	2	30143
4	Bromoviridae	18	151486
5	Caliciviridae	9	70075
6	Closteroviridae	11	182663
7	Comoviridae	17	187716
8	Dicistroviridae	11	103124
9	Flaviviridae	31	336855
10	Flexiviridae	42	316405
11	Furovirus	5	52760
12	HepatitisE-likeviruses	1	7176
13	Hordeivirus	1	10223
14	Idaeovirus	1	7681
15	Iflavirus	5	47483
16	Leviviridae	10	38908
17	Luteoviridae	16	91598
18	Narnaviridae	7	18166
19	Nidovirales	12	292492
20	Nodaviridae	8	35723
21	Pecluvirus	2	20751
22	Picornaviridae	30	230782
23	Pomovirus	4	46735
24	Potyviridae	46	453903
25	Sequiviridae	9	68920
26	Sobemovirus	9	37791
27	Tetraviridae	2	14416
28	Tobamovirus	15	96365
29	Tobravirus	3	29723
30	Togaviridae	15	173184
31	Tombusviridae	32	137821
32	Tymoviridae	11	70771
33	Umbravirus	4	16625
	Total	396	3423621

Table 2: Testing Data : unclassified ssRNA positive-strand viruses (Revised: March 5, 2004.)

	Taxonomy ID	ACCESSION	unclassified ssRNA positive-strand viruses
1	20451015	NC_003780.1	Acyrthosiphon pisum virus
2	21557557	NC_004045.1	Beet western yellows ST9 associated virus
3	11125722	NC_002604.1	Botrytis virus F
4	37595815	NC_005132.1	Botrytis virus X
5	9634052	NC_001278.1	Diaporthe ambigua RNA virus 1
6	18875445	NC_003412.1	Euprosterna elaeasa virus
7	38707888	NC_005281.1	Heterosigma akashiwo RNA virus SOG263
8	28261417	NC_004560.1	Oyster mushroom spherical virus

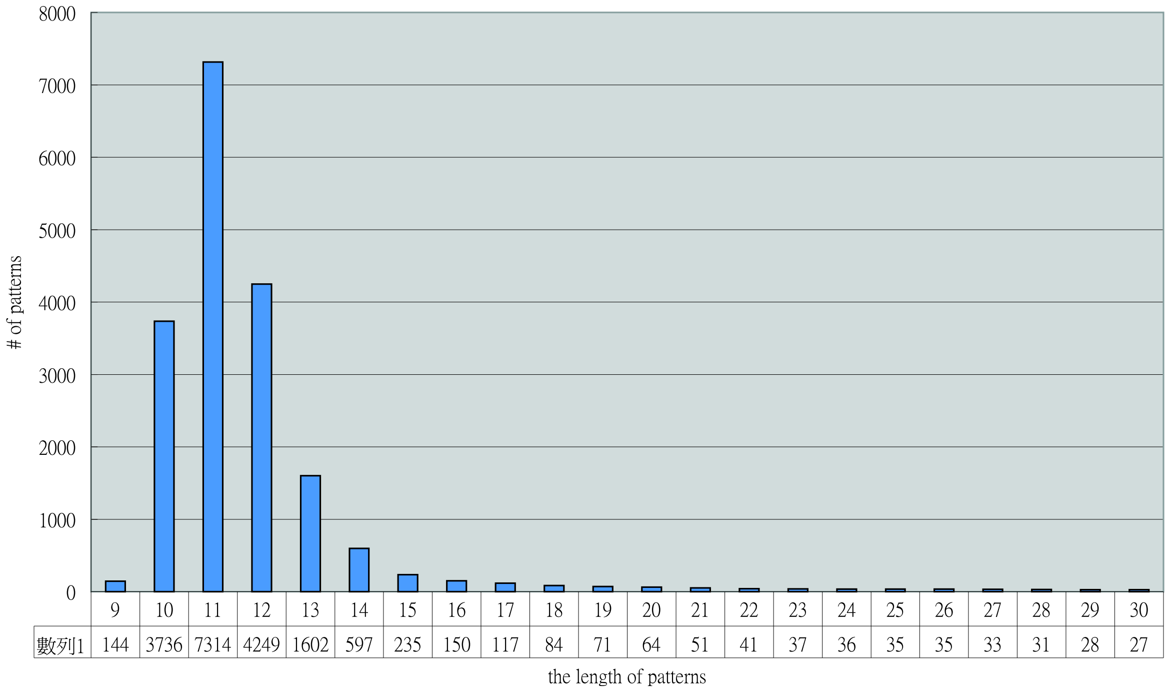

Figure 3: The distribution of the length of representative patterns

Table 3: The unknown viruses with the class achived the highest cosine similarity

	unclassified ssRNA positive-strand viruses	Class	cosine similarity
1	Acyrthosiphon pisum virus	Benyvirus	0.014302
2	Beet western yellows ST9 associated virus	Tombusviridae	0.007297
3	Botrytis virus F	HepatitisE_likeviruses	0.316871
4	Botrytis virus X	Flaviviridae	0.002858
5	Diaporthe ambigua RNA virus 1	Flaviviridae	0.003824
6	Euprosterna elaeasa virus	Flexiviridae	0.002339
7	Heterosigma akashiwo RNA virus SOG263	Potyviridae	0.006173
8	Oyster mushroom spherical virus	HepatitisE_likeviruses	0.002388

However, most of these approaches used n-grams model. On the other hand, the approach of representative patterns extraction could be used to mine for the distinct patterns for one class is interesting for the design of identifying the marker of that class.

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